

Implement multi-modal sensing system within battery pack using optical sensors and acoustic monitors to detect early signs of thermal runaway in individual cells

Executive Technology Brief

Domain: acoustic

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented recurring problem 'traction battery thermal runaway initiation' in electric vehicles: 5 of 25 NHTSA owner complaints retrieved live at benchmark time (2021–2021) involve fire or battery-related faults. Narratives: 'The high voltage battery on my 2021 Tesla Model 3 Standard Range suddenly failed without any prior warning or gradual degradation. The vehicle was consistently achieving approximately 180 miles of range per full charge. '; 'This is my fourth high voltage battery failure on this vehicle. Battery replacements occurred at approximately 13,000 miles, 60,000 miles, 96,000 miles, and now at 121,000 miles. The first failure occurred at 13,000 mile'. Energy R&D; context: 10 DOE OSTI records on thermal-runaway propagation. LIMITS STAMPED INLINE: voluntary unverified reports, causality UNVERIFIED, counts are NOT incidence (Art. XXI.5 generalized).

Mechanism: Advanced characterization techniques including X-ray imaging, optical sensing, acoustic field imaging, and spectroscopy to study thermal runaway (TR) mechanism

Architecture: PROPOSED architecture (no physical system exists): transducer element + matching layer; excitation / receive electronics; signal chain (detection criterion); coupling path to target medium — realizing: Implement multi-modal sensing system within battery pack using optical sensors and acoustic monitors to detect early signs of thermal runaway in individual cells

Governing Model: Acoustics / ultrasonics governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Key Equations:

AC-002: Wave speed relation. Expression: $c = f * \lambda$. Applicability to this invention: CONDITIONAL; holds under: non-dispersive propagation in the medium of interest. Assumptions: sound speed is medium- and temperature-dependent — source the value for the actual medium. Engaged variables: NOT ESTABLISHED.

AC-003: Exponential attenuation. Expression: $A = A_0 * \exp(-\alpha * d)$. Applicability to this invention: APPLICABLE; holds under: homogeneous lossy medium; single frequency. Assumptions: α is frequency- and medium-dependent; two-way path doubles the exponent. Engaged variables: d .

AC-001: Acoustic impedance contrast (pulse-echo). Expression: $R = ((Z_2 - Z_1) / (Z_2 + Z_1))^2$. Applicability to this invention: CONDITIONAL; holds under: normal incidence, plane wave, lossless media. Assumptions: oblique incidence and attenuation modify detectability. Engaged variables: NOT ESTABLISHED.

Kill Condition: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

Current State: ENGINEERING_DEFINITION. No physical prototype. No clinical validation.

Disclosure: This dossier distinguishes established evidence from proposals and unresolved questions.